



BrightLearn Data Analytics Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals



Research Tip: Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

An Organized Collection of data stored electronically for easy access and Management.



Example:

- ① Hospital patient records
- ② Bank account information

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

Stores large volumes of historical data from multiple sources for analysis and reporting. Unlike a regular database (handles live transactions), a warehouse handles historical analysis for strategic decision-making.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

Stores raw, unprocessed data in any format. Unlike a warehouse (clean, structured), a data lake stores everything as-is. Preferred when collecting massive diverse data for future ML or analysis.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A smaller subset of a data warehouse built for one department (e.g. marketing). Used by department-level analysts who only need relevant data.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

- * Structured → Rows and columns. e.g. Excel Spreadsheet
- * Semi-structured → Some Organisation, no rigid tables. e.g. JSON file.
- * Unstructured → No fixed format. e.g. Customer review text or video.



6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Designing new data is organised and related in a system. Prevents duplication and inconsistency.
* Conceptual model → High-level entity diagram (customer → order).

* Relational Model → tables linked by keys — used in MySQL, PostgreSQL, SQL

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

Analysing large datasets to find hidden patterns for business decisions.

Examples: Product recommendations in retail
* Fraud detection in banking.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

Software that manages databases — Create, read, update, delete data.

* MySQL: Websites and apps.

* PostgreSQL: Financial systems needing strong data integrity.

* Oracle: Large enterprises → banks, governments.

* MongoDB: Flexible, unstructured data — social media apps.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

* Extract: pull raw data from sources.

* Transform: Clean and standardise it.

* Load: Insert into a warehouse. Critical because raw data is messy and unreliable without processing.



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

- * CSV: Simple text, rows/columns. Best for small datasets and spreadsheets.
- * JSON: Key-value pairs. Best for APIs.
- * Parquet: Compressed columnar. Best for large-scale cloud analytics.
- * Avro: Binary row-based. Best for real-time data streaming.



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.
- Systems that mimic human intelligence - reasoning, learning, decision-making.
- * Narrow AI - one task only. E.g. Spam filter.
 - * General AI - Human-level reasoning across all domains. Examples: does not exist yet
- theoretical

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.
- Systems that learn from data instead of being explicitly programmed.
- * Supervised: Labelled data. Examples: spam detection.
 - * Unsupervised: Finds patterns alone. Example: customer grouping.
 - * Reinforcement: Learns via rewards/penalties.
- Examples: AI playing chess.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?
- Deep learning is ML using multi-layered neural networks. Neural networks process data through layers of connected nodes, adjusting weights based on errors until accuracy improves. Powers facial recognition, voice assistants, self-driving cars.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.
- Enables computers to understand and generate human language. Applications: voice assistants (Siri), spam filters, Google Translate.



15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

AI models trained on vast text data to handle many language tasks - unlike narrow ML models built for one task.

- * GPT-4: Writing, coding, customer support.
- * Claude: Analysis, writing, business automation - focused on safety.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

AI that creates new content (text, images, code) by learning patterns from training data.

- * Chat GPT: text generation.
- * Midjourney: Image generation.
- * GitHub Copilot: code generation.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

Data used to teach an ML model. Poor quality or biased data produces a flawed model. Example: a hiring AI trained on male-dominated historical data discriminates against women.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Enables machines to interpret images and video by processing pixel data through neural networks.
Use cases: Medical AI detecting tumours in X-rays; Amazon Go tracking products customer pick up.



Section C: APIs, Tools & Developer Concepts

Research Tip: Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?
Allows different software systems to communicate. Analogy: a waiter takes your order to the kitchen and returns with food - you never enter the kitchen. Lets developers use existing services without rebuilding them.

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

A secret token authenticating who make API requests. Store in environment variables - never in public code. If exposed, anyone can use it on your behalf, causing data breaches or financial loss.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

REST (Representational State Transfer) uses standard HTTP methods:

- * GET: Retrieve data.
- * POST: Create data.
- * PUT: Update data.
- * DELETE: Remove data.

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

JSON Lightweight text format for data exchange - readable by humans and machines. Ideal for APIs.



23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

Prompt Engineering - crafting clear, specific instructions to get the best AI output

Poor	Improved
"Tell me about data."	"Explain databases vs data warehouses for beginner."
"Write for my Business".	"Write a 3-sentence intro for a Johannesburg electrical contractor targeting corporate clients".

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

A MAP function applies an action to every item in a list.

Python example

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Delivering computing services over the internet on a pay-as-you-use basis.

- * IaaS: Virtual infrastructure. Example: AWS EC2.
- * PaaS: Development platform. Example: Google App Engine.
- * SaaS: Ready-to-use software. Example: Microsoft 365.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

Tracks file changes over time, enabling collaboration and rollback. Git is most widely used.

- * Commit: Saved snapshot of changes.
- * Branch: Separate development line for new features.



27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

Interactive environment to write code, see results immediately, and add explanatory text - all in one document. Most used with Python. Popular because it makes experimentation and sharing results fast and easy.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Simple, readable language dominant in data analytics and AI due to its vast library ecosystem.

- * Pandas: Data manipulation.
- * Numpy: Mathematics operations on arrays.
- * Matplotlib/seaborn: Data visualization
- * scikit-learn: Machine learning models.



Section D: Data Ethics, Governance & Careers

Research Tip: Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

Data Privacy is individuals' right to control their personal information. GDPR (EU:2018) gives people rights to control over their data and places strict obligations on organisations handling it. Applies to all 27 EU member states and any organisation worldwide processing EU residents' data - including South African companies with EU customers.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Moral Principles guiding responsible data use. Three unethical examples:

- * Cambridge Analytica: Harvested Facebook data without consent to manipulate elections.
- * Biased facial recognition: Higher error rates on dark-skinned faces led to wrongful arrests.
- * Price discrimination: Charging different prices based on personal data without transparency.

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

Framework of policies ensuring data is accurate, secure, and used appropriately. Key components: data ownership, quality standards, access control, Privacy compliance (POPIA in SA), data lineage tracking.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

When a model produces unfair outputs due to biased training data. Example: Amazon's hiring AI downranked women's CVs because it trained on historically male-dominated hiring data. Consequences: discrimination, wrongful decisions, legal liability, loss of public trust.



33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Data Analyst	- Interpret data for business questions Excel, SQL, Power BI, Tableau
Data Scientist	- Build predictive models Python, scikit-learn, Jupyter
Data Engineer	- Build data pipelines Spark, Kafka, Airflows, AWS
AI Engineer	- Deploy AI into production Tensorflow, PyTorch, Docker

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Tools and Processes turning raw data into insights for decisions.

- * Power BI - interactive dashboards connected to multiple data sources.
- * Tableau: Drag-and-drop visualisation for large datasets and executive reporting.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The most interesting concept was machine learning because computers can learn patterns from data and make predictions.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The most difficult concept was Data Warehouse! Understanding how all of the data gets cleaned,



Then coming back again to the data warehouse was a bit challenging.

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI will improve my future career by helping me automate tasks, analyze trends quicker, and make better. I guess it will more on survival of the fittest cause if you don't learn it, you will be left out.

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. IBM.com/cloud/learn
2. cloud.google.com/learn
3. aws.amazon.com/what-is
4. gdpr-info.eu
5. git-scm.com/doc
6. docs.python.org
7. anthropic.com
8. azure.microsoft.com/resources

Good luck! We believe in you.

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